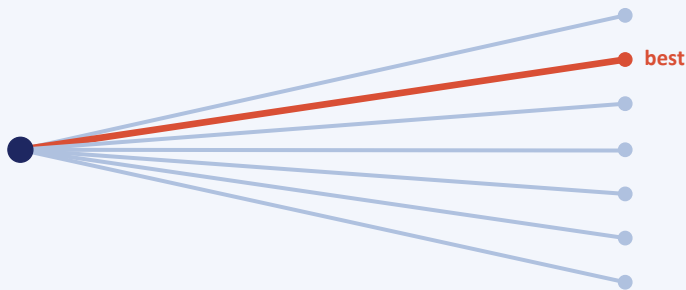


# When does thinking ahead actually help?

MSc dissertation — teaching a robot to act from a fixed dataset of recorded demonstrations, with no chance to practise

SAMPLE 50 WHOLE FUTURES, SCORE, TAKE ONE STEP



IMAGINE SHORT SEGMENTS, BRANCH, AND BRANCH AGAIN



1 **Honest conditioning is the prerequisite.** A one-shot generator can't continue from a hypothetical — force it and it invents confident nonsense the search then hunts. Made exact: **+1 subtask in 43% of trials, positive on every seed.**

2 **Search is an equaliser, not an amplifier.** Its gain grows as the generator weakens — **+9 to +37 across three generators** — and against a strong one it buys nothing at all.

3 **Nothing beat the ceiling of the data.** Across **850 trials and four methods, not one** exceeded the best behaviour the training set ever demonstrated — verified in the raw data itself.

*Not a new state of the art — a measured law of when inference-time search pays, each clause carrying its own controlled experiment.*